

Assessing the accuracy of ChatGPT responses to caregiver questions in pediatric neurology: an observational study

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Title:**Assessing the Accuracy of ChatGPT Responses to Caregiver Questions in Pediatric Neurology: An Observational Study**

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Abstract:

Background: ChatGPT has seen increased use by caregivers as a source of medical information. Pediatric neurology has added complexities because there are various disorders, changing diagnoses and potential consequences of misinformation.

Objective: To evaluate the accuracy, comprehensiveness and reproducibility of ChatGPT responses to frequently asked pediatric neurology questions.

Methods: In total, 105 unique questions across seven pediatric neurology subspecialties were included. Every question was input twice into ChatGPT (GPT-5) during separate sessions. Two certified pediatric neurologists reviewed the responses and rated them on a 4-point scale: 1- complete and accurate, 2- correct but not complete, 3- partially incorrect, 4- not correct. Reproducibility was considered to be agreement between grading categories (grade 1-2 and grade 3-4) throughout repeated responses.

Results: 60.95%(64/105) of responses were rated as comprehensive, while 31.43%(33/105) were correct but incomplete; fully incorrect responses were uncommon 3.8%(4/105). Accuracy and reproducibility were highest for structured conditions such as cerebral palsy and movement disorders, and lower for more heterogeneous domains including neurodevelopmental delay and epilepsy. Overall reproducibility was high 95.2%(100/105).

Conclusions: ChatGPT mostly delivers accurate information in pediatric neurology, with variable depth of response depending on the disorder. It is most appropriate for caregiver education with oversight by the clinician.

Introduction

The world is rapidly evolving, particularly in technology. One of the most significant 21st-century developments is artificial intelligence (AI). AI has increasingly integrated into sectors like healthcare, with significant potential to transform clinical practice and patient education (1). One of the most prominent AI applications is the Chat Generative Pre-Trained Transformer (ChatGPT), a large language model launched by OpenAI in November 2022 (2). This model is trained to mimic human language (3), and it is popular for its accessibility, ease of use, and ability to provide health-related information (4). However, relying on ChatGPT for medical advice raises accuracy concerns (5).

Several studies have evaluated ChatGPT's performance in medical contexts using validated question sets, revealing generally promising but inconsistent results. For instance, 73.4% of responses related to Dravet syndrome were rated as comprehensive (6), while an amyloidosis study found ChatGPT-3.5 achieved 89% accuracy and 79% comprehensiveness across 83 responses (7). Similarly, ChatGPT Plus performed well in the context of head and neck cancer, delivering comprehensive answers in 86.4% of cases and demonstrating 94.1% reproducibility (8). In dermatology, the model produced 88% appropriate responses, although it often struggled with complex topics (4). In the areas of cirrhosis and hepatocellular carcinoma, ChatGPT showed substantial knowledge (79.1% and 74.0%, respectively); however, fewer responses were judged to be comprehensive (47.3% and 41.1%) (9).

Further research across other medical fields has shown mixed outcomes. In bariatric surgery, 86.8% of responses were comprehensive, though accuracy declined on questions regarding recovery, risks, and complications (10). A gastrointestinal study evaluating 110 responses found treatment-related questions averaged 3.9 ± 0.8 in accuracy, while symptom-related questions scored 3.4 ± 0.8 (11). In epilepsy, ChatGPT achieved a correctness and comprehensiveness rate of 68.4% across 383 commonly asked patient questions (12). In ophthalmology, only 59.8% of answers to 115 patient-based questions were considered comprehensive (13).

Importantly, in more complex medical fields (e.g., cardiology and oncology) the overall diagnostic performance of ChatGPT was suboptimal, suggesting that further improvements are still needed to handle challenging clinical situations (14, 15). These cases indicate that while ChatGPT could be employed in an educational capacity, it is currently probably best suited to assist rather than replace professional medical advice. Although previous studies have evaluated ChatGPT in several medical specialties, pediatric neurology presents unique challenges that

warrant dedicated investigation. Many pediatric neurological disorders are rare, clinically heterogeneous, and require individualized diagnostic and management approaches that evolve with a child's development. In addition, caregivers of children with neurological conditions frequently seek health information online to better understand diagnoses, treatment options, prognoses, and long-term care. Consequently, inaccurate, incomplete, or inconsistent information generated by large language models may contribute to misconceptions, delayed medical consultation, inappropriate healthcare decisions, and increased caregiver anxiety.

Although ChatGPT is increasingly used by patients, caregivers, and medical trainees, its utility in pediatric neurology has not yet been studied. Therefore, this study is designed to evaluate overall efficiency (accuracy and reproducibility) of ChatGPT's responses to a structured set of commonly asked questions in pediatric neurology. While ChatGPT-4 generally outperformed version 3.5, none of its answers were rated significantly better (16). In contrast, our evaluation with ChatGPT-5 showed generally higher accuracy and completeness. By addressing this evidence gap, the study aims to provide meaningful insights into whether ChatGPT can be safely and responsibly integrated into pediatric neurology as a supportive tool for education and patient care.

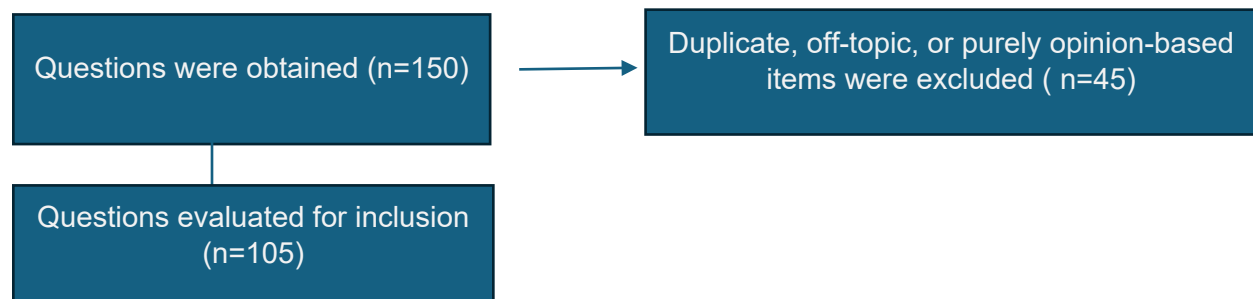
Materials and Methods

Study design

An observational study was conducted during August- November 2025. Institutional review board approval was not required for this study.

Question curation/data source

A total of 105 patient-based questions categorized in 7 domains (each with 15 questions) related to pediatric neurological conditions were obtained from Facebook patient support groups, the American Academy of Pediatrics, hospital websites, professional organizations, and health education platforms. The questions were selected to represent authentic concerns frequently raised by patients and their caregivers in pediatric neurology (Fig.1). Two pediatric neurologists independently reviewed all candidate questions for clinical relevance and clarity. Both pediatric neurologists were fully informed about the study's purpose and procedures prior to participation. Their involvement was voluntary, and informed consent was obtained from each reviewer. All contributions were treated confidentially and in accordance with relevant ethical standards. Duplicate, off-topic, or purely opinion-based items were excluded. Ambiguous or highly personalized questions were rephrased into clear, general forms while preserving their clinical intent. A record of all edits was retained for transparency. Questions were categorized by clinical domain (developmental delay, Cerebral Palsy, epilepsy, movement disorders, headaches, Pediatric Stroke and neuromuscular disorders) and by question type (definition, diagnosis, management, prognosis, or prevention). This classification enabled subgroup analyses of ChatGPT's performance across content areas.



Questions included for final analysis
(n=105)

...ually into ChatGPT (GPT-5), accessed through the
...rompt was used for all queries: “You are an experienced
pediatric neurologist. Please answer the following question clearly, accurately, and briefly for a
patient or their parents”. A new chat session was used for each question to prevent contextual
carryover. No previous conversation history or multi-turn dialogue was permitted. To assess
reproducibility, each question was submitted twice in separate sessions conducted at different
times. Responses were recorded verbatim, anonymized, and stored for blinded expert evaluation.

Question grading

Two board-certified pediatric neurologists independently graded both ChatGPT responses generated for each question. For the accuracy analysis, only the first response generated for each question was included and graded using the predefined four-level scoring system. Any disagreement between the two reviewers regarding the grade assigned to the first response was resolved through discussion until consensus was reached, and the consensus grade was used for the final accuracy analysis. Each response was rated for accuracy and completeness using a four-level scale adapted from prior ChatGPT validation studies (10, 14):

1. Comprehensive – accurate, complete, and clinically appropriate for patient use.
2. Correct but incomplete – factually correct but missing relevant details.
3. Partially incorrect – includes both accurate and inaccurate elements.
4. Incorrect – largely inaccurate or potentially misleading.

To assess reproducibility, the second response generated for each question was independently graded using the same scoring system. The grading categories assigned to the first and second responses were then compared. Responses were considered reproducible when both responses belonged to the same predefined grading category (grades 1–2 or grades 3–4); otherwise, they were classified as non-reproducible.

Statistical analysis

The data were collected and analyzed using Excel Software. Nominal variables were summarized as frequencies and percentages. Proportions of responses earning each grade were calculated. Interrater reliability between the two reviewers/coders was calculated using Cohen’s kappa statistics and demonstrated moderate agreement ($\kappa = 0.595$).

Results

A total of 105 questions were evaluated across seven neurological categories. Figure 1 represents Flow chart of pediatric neurological -related question selection. Overall, the model generated comprehensive responses for 60.95% of all questions. Performance varied across categories. The highest rates of comprehensive responses were observed in Cerebral Palsy and Movement Disorders (both 73.3%), followed by Headaches (66.7%) and Neuromuscular Disorders (60%). Moderate performance was noted in Epilepsy (53.3%) and Pediatric Stroke (53.3%), while the lowest proportion of comprehensive responses was seen in the Neurodevelopmental Delay category (46.7%) (Table 1).

Across all categories, only 3.81% of responses (4 out of 105) were rated as “completely incorrect.”. Reproducibility of responses was high overall, with consistent answers observed in

95.2% of questions (100/105). Perfect reproducibility (100%) was achieved in the Movement Disorders and Pediatric Stroke categories. In contrast, Epilepsy showed the lowest reproducibility rate at 86.6% (Table 2).

Table 1. Grading responses generated by ChatGPT to questions related to pediatric neurology categorized by question type

Category	Comprehensive	Correct but incomplete	Some correct and some incorrect	Completely incorrect
Neurodevelopmental Delay (N=15)	46.7% (7 / 15)	20.0% (3 / 15)	20.0% (3 / 15)	13.3% (2 / 15)
Cerebral Palsy (N=15)	73.3% (11 / 15)	20.0% (3 / 15)	0.0% (0 / 15)	6.7% (1 / 15)
Epilepsy (N=15)	53.3% (8 / 15)	33.3% (5 / 15)	6.7% (1 / 15)	6.7% (1 / 15)
Movement Disorders (N=15)	73.3% (11 / 15)	26.7% (4 / 15)	0.0% (0 / 15)	0.0% (0 / 15)
Headaches (N=15)	66.7% (10 / 15)	33.3% (5 / 15)	0.0% (0 / 15)	0.0% (0 / 15)
Pediatric Stroke (N=15)	53.3% (8 / 15)	46.7% (7 / 15)	0.0% (0 / 15)	0.0% (0 / 15)
Neuromuscular Disorders (N=15)	60.0% (9 / 15)	40.0% (6 / 15)	0.0% (0 / 15)	0.0% (0 / 15)
Overall accuracy(%)	60.95% (64/105)	31.43%(33/105)	3.81%(4/105)	3.81%(4/105)

ChatGPT = Chat Generative Pre-Trained Transformer, N = number.

Table 2. Proportion of questions with reproducible responses generated by ChatGPT categorized by question type.

Category	Percentage of reproducibility
Neurodevelopmental Delay (N=15)	93.3% (14/15)
Cerebral Palsy (N=15)	100% (15/15)
Epilepsy (N=15)	86.6% (13/15)
Movement Disorders (N=15)	100% (15/15)
Headaches (N=15)	93.3% (14/15)
Pediatric Stroke (N=15)	100% (15/15)
Neuromuscular Disorders (N=15)	93.3% (14/15)

Overall reproducibility	95.2% (100/105)
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ChatGPT = Chat Generative Pre-Trained Transformer, N = number.

Discussion

This study is the first to evaluate ChatGPT's responses to commonly asked, patient-based questions across multiple pediatric neurology domains, unlike previous studies that focused on a single neurological disorder. Two key findings emerged: first, response accuracy varied across different neurological domains; second, reproducibility was high in all domains, showing that the model generally produces consistent outputs.

The present study showed that ChatGPT provided complete responses for 60.95% of pediatric neurology-related questions, while only 3.8% of answers were completely incorrect. These findings imply that ChatGPT often provides clinically relevant information, although the comprehensiveness of responses varies depending on the complexity of the query (1, 2). Similar patterns of accuracy and incompleteness were also reported in previous evaluations of ChatGPT across multiple medical specialties, particularly for complex clinical topics (4, 9, 10). We found that cerebral palsy and movement disorders demonstrated the best performance, with most responses being complete and only a few errors reported. Cerebral palsy is associated with relatively stable definitions and management approaches, which could lead to more consistent and complete answers. For movement disorders, one possible explanation is that many commonly encountered movement disorder questions involve well-established clinical concepts and management principles, although the field itself remains clinically heterogeneous. In the field of ophthalmology as well, it was noticed that highly structured clinical queries performed better than large language models (14, 17, 18). By comparison, neurodevelopmental delay and epilepsy had lower rates of complete responses, likely reflective of the heterogeneity and evolving diagnostic criteria in these conditions, as well as individual treatment approaches (9, 10). Comparable trends are reported in previous investigations of ChatGPT's performance in epilepsy and other neurologically complex disorders (12, 13).

Overall, most responses were accurate or complete, indicating strong general knowledge in pediatric neurology, similar to findings in ophthalmology and bariatric surgery (10, 14). However, technical accuracy does not always ensure clinical usefulness. Although incorrect responses were uncommon, even minor gaps or inaccuracies in complex cases may have meaningful clinical consequences, particularly by affecting parental understanding and decision-making. Consistent with prior studies, these findings support using large language models as supportive tools under clinical supervision (10, 14, 17, 18).

High reproducibility was observed within specific pediatric neurology subspecialties (such as pediatric movement disorders, cerebral palsy, and pediatric stroke) with 100% reproducibility across all cases. This could be explained by the fact that these conditions are defined by clearly established criteria, consistent terminology, and quite stable diagnostic and management approaches. This structured clinical knowledge may reduce variation in generated responses, enabling the model to provide consistent information across repeated queries, as also reported in previous studies (19).

In the present study, epilepsy demonstrated the lowest reproducibility rate (86.6%), although it is higher than in a previous study about epilepsy (12), suggesting greater variability in the model's responses for epilepsy-related questions compared to other categories. This finding is clinically plausible, given the heterogeneity of epilepsy, that includes a different types seizure, etiologies, and individualized treatment strategies. Small changes in wording across responses are more likely due to incorporating different valid clinical views, not true inconsistency. Similar challenge was reported in a prior epilepsy study, particularly for prognostic related questions (12). Other domains such as neurodevelopmental delay, headaches, and neuromuscular disorders showed less than perfect reproducibility. The small differences in responses across these sessions highlight changing symptoms, developmental changes, and multiple influencing factors. In general, the reproducibility rates observed in the present study are similar to or higher than those reported in other medical fields, such as ophthalmology and oncology, supporting the reliability of large language models under controlled conditions (14, 16). Achieving similar reproducibility in pediatric neurology, a clinically complex and emotionally sensitive field, suggests that ChatGPT can provide consistent and stable information. The inter-rater agreement between the two pediatric neurologists was moderate ($\kappa = 0.595$). This likely reflects subjective differences in judging whether responses were fully comprehensive (grade 1) or factually correct but incomplete (grade 2), rather than disagreement regarding the factual correctness of the responses themselves. It is important to note that reproducibility does not necessarily imply clinical accuracy; consistent responses demonstrate model stability but do not guarantee completeness or full alignment with guideline-based clinical care, as reported in previous studies (4, 9).

This study had some limitations. The model showed variable performance across neurological domains, with better results in movement disorders than in neurodevelopmental delay and epilepsy. Moreover, performance depended on source availability, as ChatGPT cannot access subscription-based textbooks and may reflect training data bias. Additionally, the use of standardized questions may not reflect real patient language, potentially affecting responses, while the model also demonstrated limited empathy. Furthermore, question coverage was also incomplete, and outcome assessment may be subject to evaluator bias, although grading was independent and structured. Finally, regional differences in clinical guidelines may have influenced treatment-related evaluations.

Conclusion

The result of this study suggests that ChatGPT may serve as a supplementary educational resource/tool for caregivers in pediatric neurology, but it cannot replace personalized clinical consultation. Clinician interpretation remains essential to ensure accuracy, relevance, and appropriate application of AI-generated information, particularly in complex conditions where incomplete information may affect understanding and decision-making. Future studies should focus on real patient and caregiver questions, assess performance in clinical settings, and compare responses with local guidelines. Other research should aim to improve understanding and empathy in newer AI models.

Data availability

All the raw data are available upon request to corresponding author.

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Author contributions

MA: Validation, Conceptualization, Visualization, Writing – review & editing, Supervision. BA: Conceptualization, Methodology, Formal analysis, Writing – original draft, review & editing, Supervision. HA: Data curation, Conceptualization, Methodology, Formal analysis, Writing – original draft, review & editing, Supervision. RA: Writing – original draft, Data curation. SE: Writing – original draft, Data curation. SH: Supervision, Conceptualization, Writing – original draft. AA: Writing – original draft, review & editing, supervision. All authors reviewed, edited, and approved the final version of the manuscript and took responsibility for the integrity of the work.

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Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.